

SunStride Smart Station

PIEZO-SOLAR POWERED REAL-TIME BUS TRACKER & TELEMETRY HUB

Exhibition Level: Cluster Level Inter-JNV Science Exhibition

Primary Theme: Green Energy & Smart Cities

Sub-Category: IoT Embedded Systems & Sustainable Transit

Status: Official Exhibition Project Abstract

1. Executive Project Abstract

As urban centers expand rapidly worldwide, public transit infrastructure remains largely dependent on non-renewable grid electricity and lacks real-time, decentralized commuter information systems. **SunStride Smart Station** introduces a dual-harvesting green energy transit shelter integrated with an Internet of Things (IoT) monitoring framework to deliver a self-sustaining, intelligent transit node.

The project employs a hybrid renewable energy architecture combining overhead solar photovoltaic panels with piezoelectric floor transducers. Solar irradiance provides continuous baseline power during daylight hours, while commuter footsteps on piezoelectric tiles harvest kinetic pressure to supply supplemental electricity during peak traffic periods. Rather than storing energy in chemical battery banks—which introduce high costs, replacement cycles, and toxic environmental waste—the station prioritizes **direct real-time power telemetry**.

An ESP8266 NodeMCU microcontroller serves as the core processing engine. An ultrasonic proximity sensor (HC-SR04) continually monitors the bus entry bay. When an approaching vehicle crosses a calibrated **8 cm proximity threshold**:

- An active 5V audio buzzer immediately sounds an arrival chime, providing **universal accessibility** for visually impaired commuters and waiting passengers.
- The NodeMCU dynamically generates a realistic route ID (e.g., **Route #JNV-EXPRESS-204**) and updates the platform status to **BUS ARRIVED**.
- Live telemetry metrics—including solar power generation (mW), footstep kinetic power pulses (mW), cumulative energy consumed (mWh), and platform status—are broadcast continuously over Wi-Fi to a responsive web dashboard accessible on commuters' smartphones and station displays.

When the vehicle departs beyond the 8 cm limit, the system automatically silences the chime and resets to a clear platform state. **SunStride Smart Station** demonstrates how modern urban shelters can achieve self-monitoring operational efficiency, reduce municipal grid dependency, and improve transit accessibility at a minimal deployment cost.

2. Problem Statement & Research Motivation

Conventional bus stops in developing urban and rural sectors suffer from three fundamental limitations:

- Total Energy Dependency:** Municipal bus shelters rely entirely on local electrical grids for lighting and dynamic displays, incurring recurring operational costs and emitting substantial carbon footprints over time.
- Lack of Real-Time Commuter Visibility:** Waiting passengers frequently lack live arrival information, causing uncertainty, crowding, and inefficiency in public transit management.
- Accessibility Gaps:** Standard stop designs lack active audio indicators, rendering them difficult to navigate for visually impaired commuters.

SunStride Smart Station resolves these systemic challenges by uniting dual-source renewable micro-generation with accessible, budget-friendly IoT hardware.

3. Core Project Objectives



Green Energy Objectives

- Harvest ambient solar energy using photovoltaic panels for daylight baseline operations.
- Convert passenger kinetic foot pressure into supplemental electrical pulses using piezoelectric transducer matrices.
- Demonstrate *direct energy consumption telemetry* without heavy chemical battery reliance.



Smart Transit & IoT Objectives

- Automate vehicle arrival detection using ultrasonic distance scanning (< **8 cm threshold**).
- Provide real-time accessibility via an audible arrival chime for visually impaired commuters.
- Transmit live station status and energy metrics to a web-based dashboard over local Wi-Fi.

4. System Architecture & Hardware Engineering

The system architecture is structured into three distinct functional domains: *Power Generation & Telemetry*, *Central Microcontroller Processing*, and *Peripheral I/O Communication*.

Hardware Module	Specification	Electrical Parameters	Operational Function & Pin Interface
ESP8266 NodeMCU	v3 Microcontroller	3.3V / 5V DC Input	Main control unit and Wi-Fi Web Server. Manages timing, sensor sampling, and telemetry broadcast.
Ultrasonic Sensor	HC-SR04 Module	5V DC Operative	Emits 40 kHz ultrasonic pulses. Measures bus proximity in entry bay. Trig → D1 (GPIO5) Echo → D2 (GPIO4)

HARDWARE MODULE	SPECIFICATION	ELECTRICAL PARAMETERS	OPERATIONAL FUNCTION & PIN INTERFACE
Active Buzzer	5V Piezo Chime	5V DC / 20mA	Audio arrival signal activated when bus distance drops below 8 cm. Positive Signal → D5 (GPI014)
Solar Panel Array	Polycrystalline Panel	5V / 100mA Output	Captures solar irradiance to simulate/feed station power metrics. Connected to analog monitoring circuit.
Piezoelectric Discs	27mm Transducers	PZT Ceramic Discs	Arranged under walkway flooring. Generates kinetic voltage spikes under passenger footstep impact.

5. Hybrid Energy Harvesting Strategy

Complementary Solar-Piezo Mechanics: A key innovation of **SunStride Smart Station** is the complementary nature of its two energy sources. Solar panels deliver peak wattage during bright daylight hours when ambient light is abundant. Conversely, piezoelectric tiles produce energy spikes during heavy commuter foot traffic hours—such as late evening rush hours—when solar output naturally diminishes.

The Battery-Free Telemetry Approach: Conventional green energy models rely on heavy lead-acid or lithium battery banks. SunStride demonstrates direct power telemetry—monitoring raw milliwatt (mW) draw directly from solar and piezo sources and summing cumulative energy consumed in milliwatt-hours (mWh). This design eliminates chemical waste, reduces station build cost, and models modern smart-grid energy transparency.

6. Sequential Control Logic & State Machine

State Machine Operational Algorithm

1. **Initialization Phase:** NodeMCU boots up, configures GPIO pin modes, connects to local Wi-Fi / Hotspot network, and launches an HTTP server on Port 80.
2. **Proximity Measurement:** The NodeMCU triggers a 10µs HIGH pulse on the HC-SR04 **Trig** pin. It measures the high pulse duration on the **Echo** pin and calculates distance d using the formula:

$$\text{Distance } (d) = (\text{Echo Duration in } \mu\text{s} \times 0.0343 \text{ cm}/\mu\text{s}) / 2$$

3. **Conditional Threshold Evaluator ($d < 8 \text{ cm}$):**
 - **If $d < 8 \text{ cm}$ (Bus Arrived):** Drive Buzzer pin **HIGH**, set platform state to **BUS ARRIVED**, and select a randomized bus route tag (e.g., **JNV-EXPRESS-512**).
 - **If $d \geq 8 \text{ cm}$ (Bay Clear):** Drive Buzzer pin **LOW**, set platform state to **STATION CLEAR**, and clear active route tags.
4. **Telemetry Update Loop:** The microcontroller computes live solar draw (P_{solar}) and piezo footstep surge (P_{piezo}), updates cumulative energy metrics (E_{total}), and serves dynamic payloads to all connected client web browsers.

7. Universal Accessibility & Smart City Integration

Inclusive Public Transit Design: Smart cities must be accessible to all citizens. By integrating an active audio chime with real-time web telemetry, **SunStride Smart Station** creates an inclusive environment for visually impaired individuals who rely on audible cues to know when a bus has arrived in the bay.

Web-Based Commuter Information System: Commuters holding smartphones or viewing platform displays can access the station's local IP address to view real-time platform updates, incoming route numbers, and station sustainability statistics without downloading external applications.

8. Exhibition Pitch Script (1-Minute Oral Presentation)

*"Respected judges, welcome to **SunStride Smart Station**—our vision for the future of sustainable, intelligent urban transit."*

*"Traditional bus shelters are passive structures that consume non-renewable grid power and offer no real-time commuter updates. Our project transforms the humble transit shelter into a **self-monitoring, eco-harvesting smart hub**."*

"We combine dual green energy sources: overhead solar panels for daylight harvest and piezoelectric floor tiles that capture commuter footstep pressure. Powered by an ESP8266 NodeMCU microcontroller, the station detects incoming buses within an 8 cm threshold using ultrasonic sensing. Upon vehicle arrival, it immediately sounds an audio chime for visually impaired commuters, generates dynamic route IDs, and broadcasts live power telemetry to a web dashboard accessible on smartphones. Thank you!"

9. Exhibition Defense & Judges' Q&A Guide

Q1: Why incorporate piezoelectric tiles when solar panel output is generally higher?

Answer: Solar energy and piezoelectric energy are complementary. Solar power drops during cloudy weather and night hours. However, commuter foot traffic remains extremely heavy during evening rush hours. Piezoelectric harvesting captures this kinetic energy during non-solar hours, ensuring round-the-clock green generation.

Q2: Why did you choose not to include chemical storage batteries in this model?

Answer: Omitting chemical battery storage minimizes chemical waste, reduces capital costs, and allows us to demonstrate **direct real-time power telemetry**. By showing live energy draw directly on the web dashboard, the station mimics modern smart-grid monitoring where generation and consumption are tracked in real-time.

Q3: How does the system prevent false triggers caused by passing pedestrians?

Answer: The ultrasonic sensor is calibrated to a tight spatial threshold (**< 8 cm**) aligned strictly with the vehicle entry lane. Transient pedestrian movements outside this bay do not break the distance threshold, preventing false arrival alarms.

Q4: What is the real-world scalability of this project in a municipal transit network?

Answer: At scale, hundreds of SunStride stops can form a decentralized, eco-powered IoT network. Cities can lower municipal energy bills, provide real-time bus tracking to transit authorities, and improve transit equity for passengers with disabilities.

10. Conclusion & Future Scope

SunStride Smart Station successfully validates the integration of dual green energy harvesting with low-cost IoT transit automation. Future enhancements include expanding the piezoelectric flooring matrix using capacitive power smoothing, integrating RFID-based bus fleet identification, and connecting multiple smart station nodes to a centralized municipal cloud server.